

Using High Fidelity Avatars to Enhance Learning Experience in Virtual Learning Environments

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ABSTRACT

Virtual Learning Environments (VLEs) are the quintessence of the integration of several Information and Communication Technologies (ICTs) in the learning process. In this study, we investigate the performance of a multi-user Virtual Reality Learning Environment (VRLE), incorporating high-fidelity avatars, as a tool for enhancing learning experience in VLEs. This work presents the system architecture along with preliminary evaluation results provided by 19 post-graduate students.

Keywords: Virtual Learning Environments, Virtual Reality Learning Environments, Avatars, Eye Tracking, Facial Cues, Motion Capture.

Index Terms: K.3.1 [Computers and Education]: Computer Uses in Education; I.3.8 [Computer Graphics]: Applications.

1 INTRODUCTION

Virtual Learning Environments (VLEs) are nowadays commonly used by many institutions to enhance learning experience by combining several ICTs [1]. Moreover, web-conferencing services are integrated into VLEs, allowing for real-time lectures to take place between remotely located students and educators. Those services allow real-time communication, via voice over IP, along with web-camera feed and multimedia content sharing, to further increase student engagement [4], establishing VLEs as the best alternative to traditional learning [10].

Virtual Reality Learning Environments (VRLEs) is the virtual augmentation of VLE environments, which may be experienced with the use of a head-mounted display (HMD), or by a standard digital screen. VRLEs, as newly emerged applications may provide an equally beneficial alternative to VLEs in the field of education, with several studies reporting VRLEs' positive impact on the learning process [5].

An important aspect of VRLEs is their multi-user capabilities. In such environments, educators and students can control avatars and share the experience of simulated virtual worlds, increasing both student engagement and performance [6, 9]. Moreover, current commercial-oriented technologies allow real-time full-body and eye motion transference onto high fidelity avatars, whilst the potential of facial cue recognition is already under development [2, 3]. Such affordances further expand the benefits of implementing VR-based solutions into the learning process [7].

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In this work, we developed and evaluated the potential of a VRLE as a supplementary learning tool in VLEs. Our system features a 3D virtual space specifically designed to assist students in better understanding aspects of digital space and interface design, as part of their official syllabus, as well as a high-fidelity avatar representing, and motion controlled by the educator in real-time.

Students may roam around the 3D virtual world using their desktop computers, while also being able to observe the educator, who, via an avatar is giving a lecture as he normally would. Our study extends the work of Zizza et. al. [11], by featuring several additional affordances, such as transferring the educator's full-body motion, eye motion, and facial cues, from the real world onto the avatar. In this manner we exemplify how VRLEs may complement traditional learning experiences, by providing an outlet for educational content which requires hands-on perception of 3D dimensional spaces in general, but particularly understanding the configuration, design, and management of 3D digital content.

2 SYSTEM ARCHITECTURE

To develop the system presented in this work, we integrated, and customized a variety of off-the-shelf technologies. A VIVE Pro HMD¹ was used, accompanied by the two base stations typically used to track its location in the real world. Moreover, we used five (5) VIVE Trackers², to track the motion of the pelvis, hands, and feet of the educator. Combining VIVE Pro HMD and VIVE Trackers with Inverse Kinematics³, allowed us to transfer the educator's full body motion onto his avatar. The overall development of the system was carried out using Unity⁴, while the educator's avatar was created using Character Creator⁵.

In addition, ManusVR Gloves⁶ were utilized to track the educator's finger motion. Finger tracking was not optimal when using ManusVR Unity SDK, and therefore we had to author additional code to include rotation offsets and appropriately transfer finger rotation to the avatar. Finally, the Pupil Labs Eye Tracking system⁷ was utilized to transfer real eye motion onto the educator's avatar, along with the BinaryVR⁸ system, which was used for transferring facial cues via an array of BlendShapes⁹. BinaryVR's Unity SDK includes an avatar which is not compatible with any character creation software. Therefore, we had to develop a calibration tool so the educator could adjust a multiplier for the intensity of each facial cue tracked by BinaryVR, to properly match his real facial cues. Also, the educator had a mirror at his disposal to be able to check for any equipment errors which might occur during the lecture. Mirror¹⁰ was used to synchronize the data from the server, hosted at the educator's location/site, to the desktop clients that were used by students, each one at a different location also. To achieve lag-free motion of the educator's avatar, all sensor

¹ <https://www.vive.com/us/product/>

² <https://www.vive.com/us/accessory/vive-tracker/>

³ <https://docs.unity3d.com/Manual/InverseKinematics.html>

⁴ <https://unity.com/>

⁵ <https://www.reallusion.com/>

⁶ <https://manus-vr.com/>

⁷ <https://pupil-labs.com/>

⁸ <https://www.hypsense.com/>

⁹ <https://docs.unity3d.com/Manual/BlendShapes.html>

¹⁰ <https://mirror-networking.com/>

data collected from the tracking devices (628 bytes in total) had to be included in each network packet.

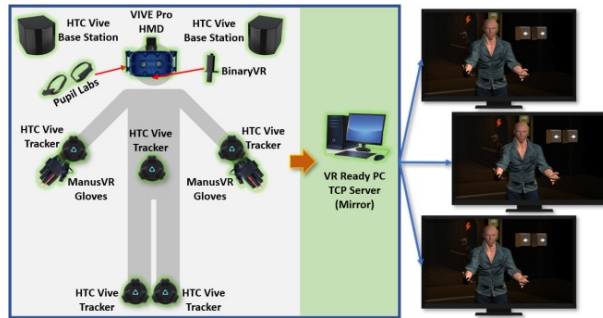


Figure 1: System Architecture.

3 USER EVALUATION

The system presented in this work was evaluated by 19 post-graduate students. The evaluation process included a web-conference supported lecture, using Big Blue Button¹¹ (BBB), with the lecture's topic – and in accordance with the actual syllabus – being User Interfaces (UI) in games [8] featuring integrated multimedia content. Following the BBB-based lecture, students were prompted to download and execute the client application. From then on, they could use their keyboard to roam around a game-like virtual world used as a testbed. Each student also had an avatar representation, however in order to not draw attention from the educator those were not anthropomorphic. The educator, through his avatar, presented the spatial arrangement of digital content and UI concepts in games by pointing to actual elements of the virtual space, and discussing them with students when required. Students were able to move around the digital space and examine spatial arrangement of digital content and their associated concepts in UIs in games. Once the VRLE session was completed, students were asked to fill an online survey evaluating their learning experience.



Figure 2: Student's view of the lecture taking place in the VRLE featuring a 3D virtual world and the educator's high-fidelity avatar.

The VRLE lecture's duration was ~13 minutes, the average internet connection speed of the participants was 23 Mbps, average Round-trip delay (RTT) was 50ms, and the average performance of the application was 148 frames per second (FPS). 95% of the participants confirmed the synchronization of the avatar's motion, facial cues and eyes motion with the instructor's voice and detected no latency. However, 20% of the participants had difficulties on following the avatar's facial cues and eyes motion due to their PCs' small screen size, while 30% experienced latency at some point during the lecture. However, those participants had either slow internet connections (<10 Mbps) or experienced low FPS due to

using low-end PCs. All students (100%) considered the VRLE experience to be beneficial and a useful supplementary learning tool adjacent to the original VLE lecture. Finally, the full body and eyes motion along with the facial cues of the instructor's avatar, assisted 90% of the students into maintaining their attention during the lecture therefore increasing their understanding towards the concepts presented.

4 CONCLUSION & FUTURE WORK

This work presents the system architecture and preliminary evaluation results of a VRLE, integrating a high-fidelity avatar which features facial cues, eye and full-body motion, recorded from an educator in real time using a variety of technologies. Preliminary evaluation results highlighted the system's usability along with its potential to enhance the learning experience in VLEs when used as a supplementary tool. Broader evaluation trials with a greater number of participants, variant educational modules, as well as additional evaluation tools will further assist into verifying these positive results for the generation of future VRLEs.

ACKNOWLEDGEMENT

The research work has been supported by the Hellenic Foundation for Research and Innovation (H.F.R.I.) under the "First Call for H.F.R.I. Research Projects to support Faculty members and Researchers and the procurement of high-cost research equipment grant" (Project Number: HFRI-FM17-1168).

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¹¹ <https://bigbluebutton.org/>